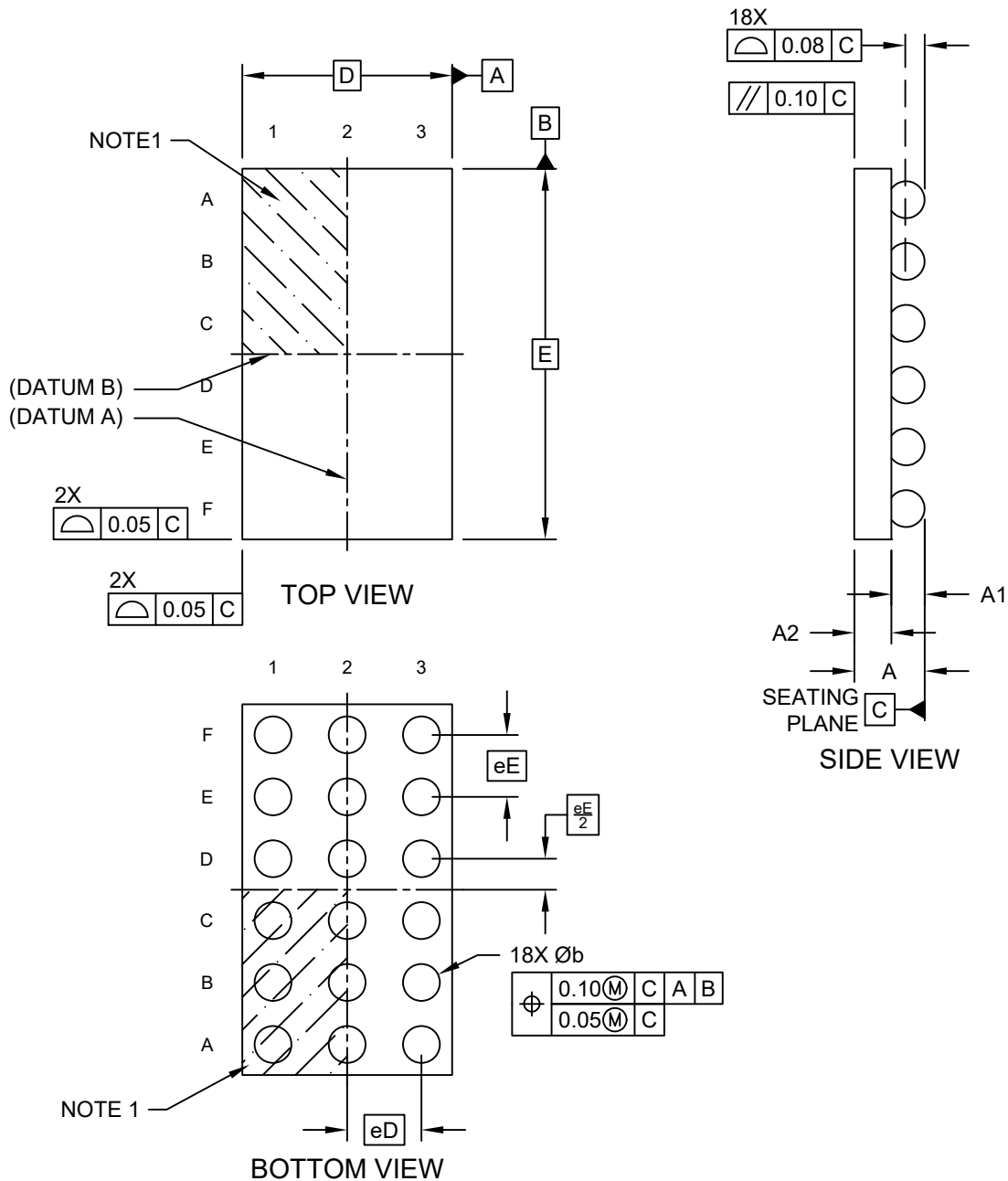


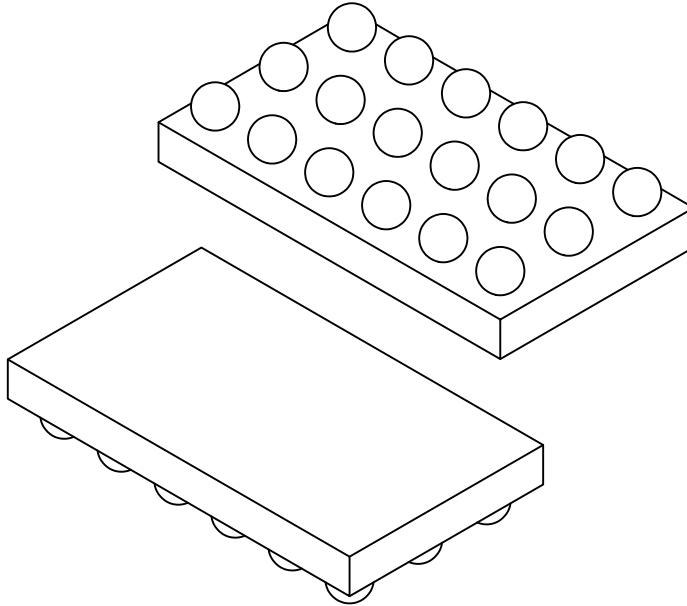
18-Ball Wafer Level Chip Scale Package (CS) - [WLCSP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



18-Ball Wafer Level Chip Scale Package (CS) - [WLCSP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



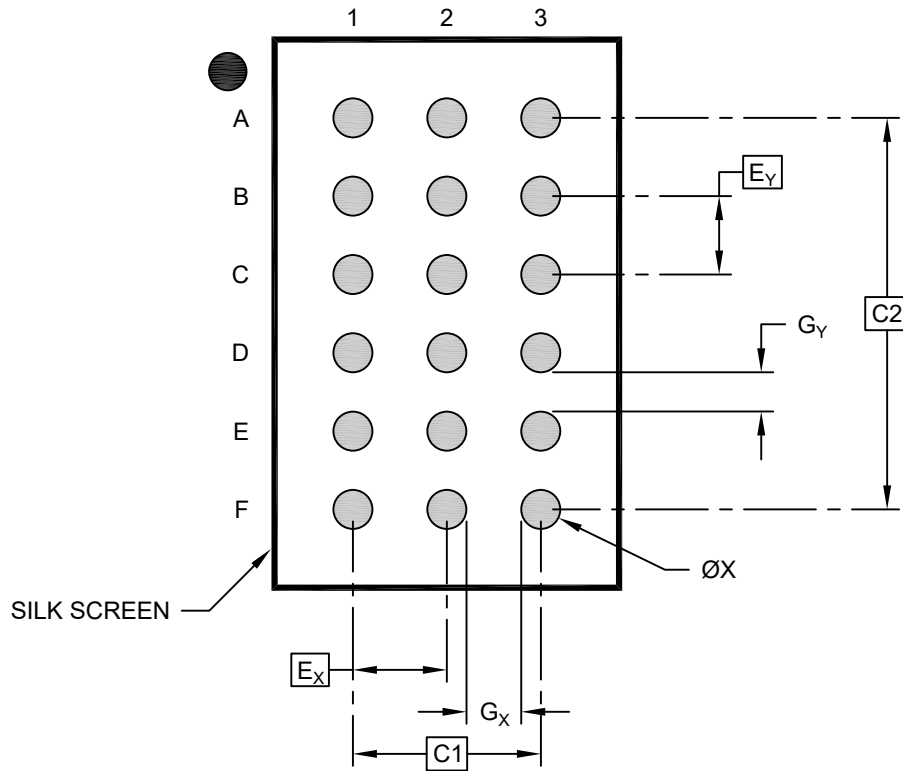
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	18		
Pitch	eD	0.60 BSC		
Pitch	eE	0.50 BSC		
Overall Height	A	0.496	0.532	0.568
Standoff	A1	0.204	0.227	0.250
Die Thickness	A2	0.292	0.305	0.318
Overall Length	D	NOTE 4		
Overall Width	E	NOTE 4		
Terminal Width	b	0.277	0.326	0.375

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
4. Package size varies with specific devices. Please contact your local Microchip representative for specific details

18-Ball Wafer Level Chip Scale Package (CS) - [WLCSP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension	Units	MILLIMETERS		
	Limits	MIN	NOM	MAX
Contact Pitch X-Axis	E_X		0.60 BSC	
Contact Pitch Y-Axis	E_Y		0.50 BSC	
Contact Pad Spacing	C1		1.20 BSC	
Contact Pad Spacing	C2		2.50 BSC	
Contact Pad Width (X18)	X1			0.25
Contact Pad to Contact Pad	G_X	0.35		
Contact Pad to Contact Pad	G_Y	0.25		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-8037-CS Rev A